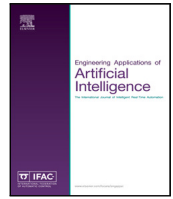




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Research paper

Low-light image enhancement using dual cross attention

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ABSTRACT

Low-light image enhancement (LLIE) aims to improve the perceptibility and interpretability of images captured in poorly illuminated environments. Existing LLIE methods often fail to capture the local self-similarity and long-range dependencies at the same time, causing the loss of complementary information between multiple modules or network layers, ultimately resulting in the loss of image details. To alleviate this issue, we design a hierarchical mutual Enhancement via a dual cross-attention transformer (ECAFormer), which introduces an architecture that enables concurrent propagation and interaction of multiple disentangling features. To capture the local self-similarity, we design a Dual Multi-head Self-Attention (DMSA), which leverages the disentangled visual and semantic features across different scales, allowing them to guide and complement each other. Further, a cross-scale DMSA block is incorporated to capture residual connections, thereby integrating cross-layer information and capturing the long-range dependencies. Experimental results show that the ECAFormer reaches competitive performance across multiple benchmarks, yielding nearly a 3.7% improvement in Peak Signal-to-Noise Ratio (PSNR) over the suboptimal method, demonstrating the effectiveness of information interaction in LLIE. For facilitating the efforts to replicate our results, our implementation is available on GitHub¹

1. Introduction

Capturing images in low-light conditions presents significant challenges in computer vision, including the degradation of fine details, reduced color saturation, diminished contrast and dynamic range, and uneven exposure. These issues can greatly compromise the integrity and clarity of captured data, negatively impacting the effectiveness of subsequent applications such as autonomous driving systems (Ranft and Stiller, 2016; Cai et al., 2020; Bresson et al., 2017) and nighttime surveillance efforts (Fu et al., 2021; Buluswar and Draper, 1998). Naturally, enhancing the visibility of objects and details in low-light images is crucial, possessing far-reaching implications for a spectrum of vision applications.

Currently, efforts for LLIE have predominantly utilized deep-learning techniques, which can be categorized into three main approaches: Conventional, Guided, and Two-Branch Methods, as depicted in Fig. 1. Conventional methods, such as the Low-light Net (LLNet) (Lore et al., 2017) and U-shaped Transformer (UFormer) (Wang et al., 2022), hinge on data-driven strategies and tailored loss functions but are constrained by data complexity, hindering the model's ability to capture robust representations. To overcome this, guided methods (Zhang et al.,

2023, 2024) integrate prior knowledge into LLIE. They decompose the input into essential and target components, leveraging the former knowledge to guide the enhancement of the latter through specialized networks. For example, some guided methods (Zhuang et al., 2021; Li et al., 2023) based on Retinex theory (Land and McCann, 1971) focus on preserving the consistency of the reflection component while adjusting the illumination. However, guided methods introduce ideal bias along with the guidance, which affects the performance of the model. Another method for incorporating prior knowledge is the Two-Branch Methods, which involve refining the input by dividing it into separate sub-components and processing each component with its dedicated enhancement network (Ma et al., 2021; Tang et al., 2022; Aherrahrou and Tairi, 2023; He et al., 2023). However, despite the use of prior knowledge, these methods fail to capture the local self-similarity and long-range dependencies. This limitation, inherent in the model architecture, can result in a loss of image details during enhancement.

Regarding the current situation in the LLIE field, we focus on enhancing the information interactions across different disentangled components and layers to fill in the image details. To achieve this, we

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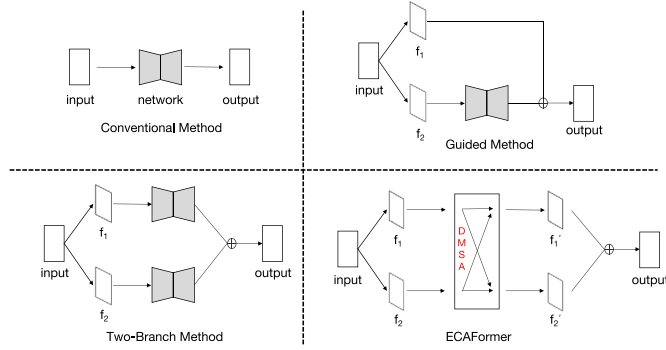


Fig. 1. Main architecture comparison. Compared to other methods, our approach utilizes DMSA to simultaneously propagate two features forward and facilitate interactions at different scales. This method is advantageous for extracting latent connections between features.

employ the U-shaped architecture as the foundation of our model to enhance low-light images. Specifically, we proposed ECAFormer as shown in Fig. 2. Following the idea of two branch methods, we incorporate a knowledge-driven strategy that disentangles the images into distinct visual and semantic components, which facilitates detail preserving with the guidance of semantic information. To capture local similarities, we emphasize leveraging the interactive advantages of both visual and semantic components by integrating a modified transformer that effectively harnesses these elements. Specifically, to enhance the interactions across different sub-components in the latent space, we designed DMSA to handle multiple inputs, which captured potential relationships between different components. Notably, the transition from Multi-Head Self-Attention (MHSA) in the vanilla transformer to DMSA modules is strategically designed to simultaneously exploit both long- and short-range dependencies within the image data. Furthermore, to capture long-range dependencies, we designed a Cross-Scale DMSA (CSDMSA) that facilitates the interaction between residual and current layer information. This mechanism is ingeniously leveraged to refine the residual connections within ECAFormer. Our main contribution can be summarized as follows:

- We explore a novel LLIE method named ECAFormer, which illustrates the importance of information integration across layers and components for detail preservation in LLIE.
- We collect a novel traffic image dataset (Traffic-297), which includes various traffic scenes and provides a new benchmark in LLIE.
- We achieve competitive results on five LLIE datasets, emphasizing computational frugality and parameter parsimony.

The paper is structured as follows. In Section 2, we respectively reviewed the related work in the field of low light enhancement and cross-attention mechanisms. In Section 3, we introduced the problem definition, the proposed method, and the details of model optimization. In Section 4, we compared our model with other methods using six publicly available datasets and conducted ablation experiments to validate its effectiveness. Finally, we summarized in Section 5 and proposed directions for future work.

2. Related work

2.1. Convolutional neural networks based LLIE

The rapid advancement in low-light image dataset collection has led to the emergence of numerous deep learning-based enhancement methods (Li et al., 2021, 2025). LLNet (Lore et al., 2017) introduced a variant of the stacked sparse denoising autoencoder for improving

degraded images, establishing a foundational framework for the application of deep learning in image enhancement. A retinex based network (RetinexNet) (Wei et al., 2018) employed a deep Retinex-based architecture to enhance low-light images by decomposing them into illumination and reflectance components. Retinex-inspired Unrolling with Architecture Search (RUAS) (Liu et al., 2021) utilized an advanced search unfolding technique based on a Retinex architecture. A highly effective unsupervised generative adversarial network (EnlightenGAN) (Jiang et al., 2021) innovatively used a generative inverse network as the primary framework, initially training with unpaired images. Joint Low-light Enhancement and Deblurring in the Dark (LED-Net) (Zhou et al., 2022), is a robust network specifically designed to address the dual challenges of low-light enhancement and deblurring simultaneously. Lightweight Attention-guided ConvNeXt Network (LACN) (Fan et al., 2023) proposed a novel module by introducing the attention mechanism into the ConvNeXt backbone network.

However, these Convolutional Neural Networks (CNN) based models fail to capture long-range dependencies, resulting in the loss of detailed information. To alleviate this, this paper introduces the transformer architecture and specifically designed cross-scale DMSA block, which significantly enhances the model's ability to extract long-range dependencies and global information.

2.2. Disentangling based LLIE

Compared to conventional LLIE methods (Liu et al., 2024; Zhu et al., 2023; Chen et al., 2024), the disentangling-based techniques (Baysal et al., 2021; Khodadadi et al., 2024; Kalita et al., 2024) have more effectively captured the latent knowledge within the data and significantly enhanced model performance. Among these approaches, it can be classified into two types, as illustrated in Fig. 1: (1) Two-Branch Methods, which optimize distinct features using separate networks, such as Context-Sensitive Decomposition Generative Adversarial Network (CSDGAN) (Ma et al., 2021) and flexible Low-Light Image Enhancement via Disentangled Representations (DRLIE) (Tang et al., 2022); (2) Guided Methods, which use one feature to guide the optimization of another feature, such as Signal-to-Noise-Ratio aware transformer (SNRNet) (Xu et al., 2022) and retinex-based transformer (RetinexFormer) (Cai et al., 2023).

However, the aforementioned model architectures inadequately exploit the complementary information among the decomposed sub-components, frequently addressing them in isolation. This approach hinders the network's ability to learn the local self-similarity and the intrinsic relationships between multiple sub-components. Therefore, we propose the DMSA module, which can simultaneously handle and inter-relate two features during forward computation. This modification enables the model to learn the relationships between sub-components, integrating their respective advantages, thus effectively capturing the non-local self-similarity and the intrinsic relationships to improve the final output of the model.

3. Methodology

In this section, we present a comprehensive description of each network module proposed and the corresponding loss functions employed. Inspired by the successful U-shaped architecture in LLIE (Li et al., 2021), we introduce a transformer network with a U-shaped architecture, depicted in Fig. 2. Specifically, ECAFormer comprises three main components: (1) Visual-semantic convolution module 3.2.2, which generates short-range features and long-range features. (2) The U-shaped cross attention Transformer 3.2.3 interacts with long- and short-term input features through DMSA, simultaneously propagating these features. (3) Mapping convolution, where the module projects the interacted features back to image features $\in \mathbb{R}^{C \times H \times W}$. In practice, the low-light input image undergoes convolution filtering to extract semantic information and visual detail information respectively. The enhanced image is then obtained through interaction enhancement of a U-shaped network using semantic and visual features.

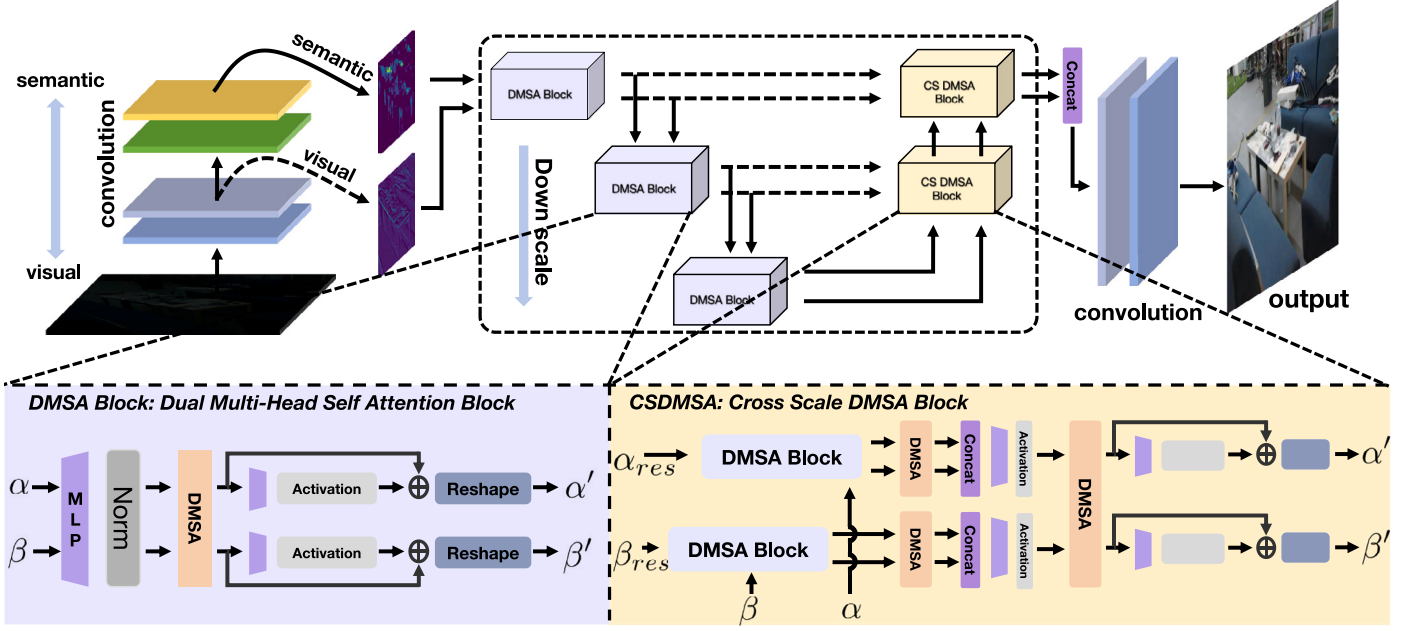


Fig. 2. The flowchart of ECAformer mainly consists of three parts: (1) Visual-Semantic Convolution modules 3.2.2, which output short-range features (visual-features) and long-range features (semantic-features). (2) The U-shaped cross-attention Transformer 3.2.3 engages with long- and short-term input features through DMSA, concurrently propagating these features. (3) Mapping convolution, where the module projects the interacted features back to image features. $\in \mathbb{R}^{C \times H \times W}$.

3.1. Problem definition

Low-light image enhancement aims to mitigate the challenges associated with low brightness, low contrast, noise, and artifacts in images captured under inadequate lighting conditions. These issues arise from the loss of visual information under insufficient illumination or complex environmental factors at the time of image capture. The objective of low-light image enhancement is to utilize models to restore and enhance low-light images $\mathbb{L}$, bringing them closer to normal images $\mathbb{N}$ and rendering them more perceptible to the human visual system.

The problem can be mathematically modeled as follows:

$$\hat{I} = f(I, \theta), \quad (1)$$

where $I \in \mathbb{R}^{C \times H \times W}$ represents the input low-light images, $\hat{I} \in \mathbb{R}^{C \times H \times W}$ represents the enhanced images, and θ represents the parameters of the network function f .

The optimization objective is to find the optimal parameters θ that minimize the differences between the enhanced images and the reference high-light or normal images. This can be formulated as:

$$\hat{\theta} = \arg \min_{\theta} \ell(\hat{I}, R), \quad (2)$$

where $R \in \mathbb{R}^{C \times H \times W}$ denotes the well-exposed reference or normal images, and ℓ denotes the loss function used to guide the network.

3.2. Model framework

3.2.1. Preliminary

In LLIE, CNNs are adept at extracting intricate local features, while Transformer networks excel in capturing valuable global feature information from complex environments. As the depth of the convolution layers increases in CNNs, the extracted features progressively embody richer semantic information. Consequently, features obtained from initial shallow convolution operations are predominantly visual, while those derived from deeper convolution layers contain more sophisticated semantic insights. To capitalize on these characteristics, we have developed a decoupling convolution extractor, which decouples the image to the visual and semantic features.

$$\begin{aligned} f_v &= \text{Conv}_1(I), \\ f_s &= \text{Conv}_2(f_v), \end{aligned} \quad (3)$$

where f_v retains an abundance of detailed visual features, while f_s preserves advanced semantic features. Both attributes are pivotal for achieving the enhanced final result.

The attention mechanism facilitates interactivity among elements, significantly enhancing global feature extraction capabilities. The conventional self-attention mechanism is defined by Eq. (4), where the vectors Q , K , and V represent query, key, and value derived from a single input. This operation selectively focuses on crucial information, optimizing resource utilization and swiftly capturing the most relevant data. Leveraging this advantage, we have developed a unique DMSA to facilitate the fusion of two distinct features. This module processes two inputs and enables their interaction via the attention mechanism while preserving their dimensional integrity. A detailed discussion of this module will be provided later.

$$\text{MHSA}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V. \quad (4)$$

We have developed a U-shaped network structure with enhanced residual information, specifically tailored to improve the network's capacity for multi-scale interaction. This architecture facilitates a comprehensive synthesis of features across various scales. During the down-sampling phase, we strategically employed a $2\times$ scaling factor to compel the network to undergo a rigorous compression process, as we separate the image into visual and semantic features. This intentional compression is crucial as it allows the network to extract more refined global information, which is essential for understanding broader contextual cues. The methodology and details of the down-sampling stage are illustrated in Eq. (5).

$$[f_v, f_s]^{(i)} = \text{RS}(\text{DMSA}_i([f_v, f_s]^{(i-1)})), \quad (5)$$

where $\{i \in 1, 2\}$ denotes the step of the down-sample stage, $\text{RS}(\cdot)$ represents the resample operation, the DMSA_i denotes the DMSA Block varies at i stages. At the bottom of the U-shaped network, we used a bottleneck consisting of two DMSA modules. And in the up-sampling process, we employed operations proportional to down-sampling. During the up-sampling phase, we utilized CSDMSA to effectively retain and restore intricate details that are often lost during the down-sampling process through residual information. The residual information initially engages in cross attention with their corresponding features, followed by

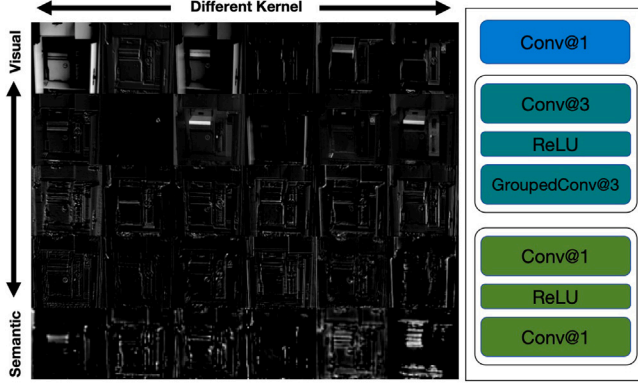


Fig. 3. From top to bottom are the shallow outputs and deep outputs of the model, respectively.

cross attention between the two types of features in the CSDMSA block. This process is crucial in maintaining the fidelity of feature representations, ensuring that the reconstructed outputs closely resemble the original inputs. Finally, we obtain the final output through concatenation and mapping convolution. It can be represented by Eq. (6).

$$\hat{I} = \mathcal{A}_{agg} (f_v^{(T)}, f_s^{(T)}). \quad (6)$$

3.2.2. Visual-semantic convolution module

Inspired by the potent capability of convolution layers to enhance local features, we devised a visual-semantic convolution module specifically designed to capture local characteristics. In CNNs, as the number of convolution layers increases, the receptive field of the model progressively enlarges, as illustrated in Eq. (7), leading to the extraction of increasingly complex semantic features.

$$l_i = l_{i-1} + \left[(k_i - 1) \cdot \prod_{j=1}^{i-1} s_j \right], \quad (7)$$

where l_{i-1} is the receptive field size of the previous layer, k_i is the kernel size or pooling window size at layer i , s_j is the stride of layer j .

As shown in Fig. 3, it can be observed that visual feature f_v focuses more on fine details, while semantic feature f_s emphasizes broader connections and contextual relationships within the image. However, this expansion often results in the attenuation of fine-grained detail within the features. To address this, we introduce a dedicated convolution module that outputs two distinct types of features: f_v and f_s . Here, f_v is derived from shallow convolution layers, capturing detailed visual features, whereas f_s emanates from deeper convolution layers, encapsulating higher-level semantic information. Specifically, in this paper, the output feature of the 2_{nd} convolutional layer is taken as f_v , and the output feature of the 5_{th} layer as f_s . Additionally, the CNNs can capture periodic and local spatial features, addressing the spatial context induction bias in transformers that rely solely on positional embedding. We employ depth-wise separable convolution within this module to enhance the rate of forward propagation without compromising much accuracy. Upon processing through this module, the network then employs attention mechanisms to facilitate a dynamic interaction between these two distinct features.

3.2.3. DMSA

Self-attention is highly efficient at directing concentrated attention toward pivotal information, thereby optimizing resource utilization and rapidly acquiring the most relevant data. However, its standard configuration is restricted to processing only a single input, which can limit its applicability in scenarios requiring complex interactions between multiple data streams. To address this constraint, we have

innovated the DMSA module as shown in Fig. 4, a sophisticated mechanism designed to effectively manage and integrate dual inputs. In this module, inputs α and β are processed to generate distinct sets of Q , K , and V vectors through separate mappings. We strategically cross the key vectors of both inputs within this module to enhance their inter-connectivity, fostering a richer, more comprehensive interaction. This is achieved by the multiplication of the query vector with the corresponding crossed key vector as in Eq. (8), producing an attention map that encapsulates the intricate dynamics between the two inputs. Enhanced by the adaptive sampling operations of the U-shaped network, this setup facilitates a profound interaction across multiple scales, allowing for a nuanced integration of features that is essential for analysis and interpretation tasks.

$$\alpha' = \text{CrossAttn}(Q_\alpha, K_\beta, V_\alpha) = \text{softmax}(Q_\alpha K_\beta^T * \zeta) V_\alpha, \quad (8)$$

$$Q_\alpha = \alpha * W_q, K_\beta = \beta * W_k, V_\alpha = \alpha * W_v.$$

The scaling factor ζ is determined through optimization. β is obtained using a similar method. The DMSA module simultaneously accepts both α and β . During the forward pass, their respective keys are exchanged, and similarity calculations are performed between each one's own query and the key from the other. This information is then used for weighted aggregation of their own value, enabling effective self-enhancement by leveraging the other's information. Unlike methods that exchange values, this approach ensures that they can enhance themselves using the other's features, while remaining distinct and not blending with each other. To preserve the spatial positional relationships among pixels, we have incorporated a position embedding (*PosEmb*) module, executed via convolution operations. Consequently, the output of the Dual Multi-head Self Attention module $[\alpha', \beta']$ can be formally defined in Eq. (9).

$$[\alpha', \beta'] = [\text{DMSA}(Q_\alpha, K_\beta, V_\alpha) + \text{PosEmb}(V_\alpha), \quad (9)$$

$$\text{DMSA}(Q_\beta, K_\alpha, V_\beta) + \text{PosEmb}(V_\beta)].$$

3.2.4. CSDMSA

During the downsampling process, detail loss often leads to image blurring, making the utilization of residual information crucial. To address this, we employ a specially designed Cross-Scale Dual Multi-head Self Attention (CSDMSA) module to handle cross-layer residual information, aiming to reduce image degradation.

First, the residual and intermediate features are processed by the DMSA module as follows:

$$[\alpha'_{res}, \alpha'_{mid}] = \text{DMSA}([\alpha_{res}, \alpha_{mid}]), \quad (10)$$

$$[\beta'_{res}, \beta'_{mid}] = \text{DMSA}([\beta_{res}, \beta_{mid}]),$$

where α_{mid} and β_{mid} represent the features of the current layer, while α_{res} and β_{res} correspond to the residual information, respectively. Next, we concatenate the corresponding features to facilitate subsequent interaction operations:

$$\alpha_{agg} = W \cdot \text{concat}([\alpha'_{res}, \alpha'_{mid}]) + B, \quad (11)$$

$$\beta_{agg} = W \cdot \text{concat}([\beta'_{res}, \beta'_{mid}]) + B,$$

where α_{agg} and β_{agg} represent the features after fusion, W is the learned weight matrix and B is the learned bias matrix. Finally, the interaction between α_{agg} and β_{agg} is processed through another DMSA module, followed by a resampling operation. The final output $[\alpha', \beta']$ of the CSDMSA module is given by:

$$[\alpha', \beta'] = \text{CSDMSA}([\alpha, \beta]) \quad (12)$$

$$= \text{RS}(\text{DMSA}([\alpha_{agg}, \beta_{agg}])),$$

where $\text{RS}(\cdot)$ denotes the resampling operation.

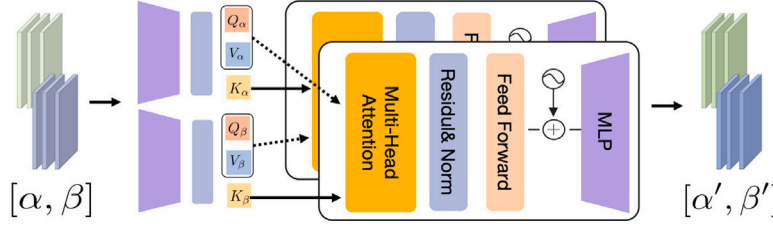


Fig. 4. DMSA: A highly symmetrical module is depicted in the figure, which illustrates the process of computing the $[\alpha', \beta']$ guided by each other using cross attention.

3.3. Loss function

We incorporated two types of loss functions that better align with human visual perception and facilitate faster model training. The Total Loss $\mathcal{L}_{Total}$ can be represented as Eq. (13), $\lambda \in [0, 1]$.

$$\mathcal{L}_{Total} = \lambda * \mathcal{L}_p + (1 - \lambda) * \mathcal{L}_c, \quad (13)$$

where $\mathcal{L}_p$ is the loss of perceptual and $\mathcal{L}_c$ is the loss of charbonnier.

3.3.1. Perceptual loss

Perceptual Loss (Johnson et al., 2016) adopts an efficient approach by quantifying the discrepancies through the squared error between features extracted from specific layers or an aggregation of multiple layers after both the ground truth and the reconstructed image have traversed the same pre-trained neural network. The Perceptual Loss $\mathcal{L}_p$ is given as follows:

$$\mathcal{L}_p = \sum_{i=1}^n \frac{1}{C_i H_i W_i} \left| F_i^l(f(I)) - F_i^l(R) \right|^2. \quad (14)$$

Among them, f symbolizes the enhancement network. F_i^l indicates the i th feature map in the l th layer. We have utilized a VGG-19 network pre-trained on ImageNet and employed the output feature maps from its initial five ReLU layers to compute the loss. This method leverages the deep network's architecture to extract rich, complex feature representations that are critical for assessing the perceptual quality of the enhanced images.

3.3.2. Charbonnier loss

Compared to the conventional L_1 Loss, Charbonnier Loss exhibits superior robustness and stability, particularly when dealing with outliers. Additionally, the computational efficiency of Charbonnier Loss is enhanced due to its reliance on a single square and root operation, in contrast to the ordinary L_1 Loss which necessitates an absolute value computation. This streamlined calculation speeds up the processing and contributes to smoother gradients, facilitating more effective optimization during model training. The Charbonnier Loss $\mathcal{L}_c$ is given as follows:

$$\mathcal{L}_c = \sum_{i=1}^n \sqrt{(\hat{I} - R)^2 + \epsilon^2}. \quad (15)$$

Benefiting from the addition of ϵ , the phenomenon of gradient vanishing when $\hat{I}$ and R are very close has been alleviated, making the model easier to train.

4. Experiment

4.1. Experimental configurations

4.1.1. Datasets

We conducted comparisons across multiple public datasets: LOL-v1 (Wei et al., 2018), LOL-v2 (Yang et al., 2021b), SID (Chen et al., 2019) and SMID (Chen et al., 2018). Additionally, we have collected the *Traffic-297* dataset, which includes various traffic scenes. It contains an amount of real nighttime traffic scenes with complex lighting environments, which pose a significant challenge to low-light enhancement models.

Table 1

Comparison of PSNR of different methods across 5 public datasets. The data was obtained either by training with publicly available code or from the information provided in the paper. ECAFormer achieved promising results without a significant increase in parameters.

Methods	LOL-v1	LOL-v2-r	LOL-v2-s	SID	SMID	AVG	Params
RN(BMVC 2018)	16.77	15.47	17.13	16.48	22.83	17.74	0.84
KinD(ACMMM 2019)	20.86	14.74	13.29	18.02	22.18	17.82	8.02
EnGAN(TIP 2021)	17.48	18.23	16.57	17.23	22.62	18.43	114.35
RUAS(CVPR 2021)	18.23	18.37	16.55	18.44	25.88	19.49	0.00
DRBN(TIP 2021)	20.13	20.29	23.22	19.02	26.60	21.85	5.27
UF(CVPR 2022)	16.36	18.82	19.66	18.54	27.20	20.12	5.29
RSTM(CVPR 2022)	22.43	19.94	21.41	22.27	26.97	22.60	26.13
MIRNet(TPAMI 2022)	24.14	20.02	21.94	20.84	25.66	22.52	31.76
SNR(CVPR 2022)	24.61	<u>21.48</u>	24.14	22.87	<u>28.49</u>	<u>24.32</u>	4.01
LLF(AAAI 2023)	23.65	20.06	24.04	\	\	\	24.55
RF(ICCV 2023)	23.69	21.43	<u>24.16</u>	<u>23.85</u>	27.55	24.14	1.61
U2E-Net(PR 2024)	23.92	\	\	\	\	\	\
ECAFormer	<u>24.24</u>	21.99	25.86	24.69	29.34	25.22	2.50

Table 2

Parameter adjustment formulas based on variables F , Y , and Z .

Parameter	Value
Exposure	$-5 + 5F$
Highlights	$50 \min\{Y, 0.5\} + 75$
Shadows	$-100 \min\{Z, 0.5\}$
Vibrance	$-75 + 75F$
Whites	$16(5 - 5F)$

The *Traffic-297* dataset was collected from traffic checkpoint surveillance cameras, covering a wide range of real-world scenarios. It includes images captured under various weather conditions (e.g., clear, rainy, foggy) and at different times of day, including both daytime and nighttime scenes. Notably, nighttime images contain diverse lighting sources such as vehicle headlights, street lamps, and illumination from distant buildings, leading to complex lighting conditions.

From the initially collected 2000 raw images, we filtered out visually redundant or overly simple scenes, retaining 297 images that exhibit high diversity in scene composition and illumination conditions. Among these, 247 images are used for training and the remaining 50 for testing.

To construct paired training data, we followed the same image generation methodology as in the LOL dataset, where high-light and low-light image pairs are synthesized from daytime captures. Our model is then evaluated on real nighttime scenes to verify generalization. Detailed acquisition settings and preprocessing parameters are provided in Table 2.

4.1.2. Implementation details

We implemented the ECAFormer model using PyTorch. The model was trained on an NVIDIA RTX 4090 with 24 GB of VRAM using the Adam optimizer ($\beta_1 = 0.9$ and $\beta_2 = 0.999$), across a total of 250,000 iterations. The initial learning rate was set at 2×10^{-4} and gradually reduced to 1×10^{-6} through a cosine annealing schedule during the training process. Training samples were generated by randomly

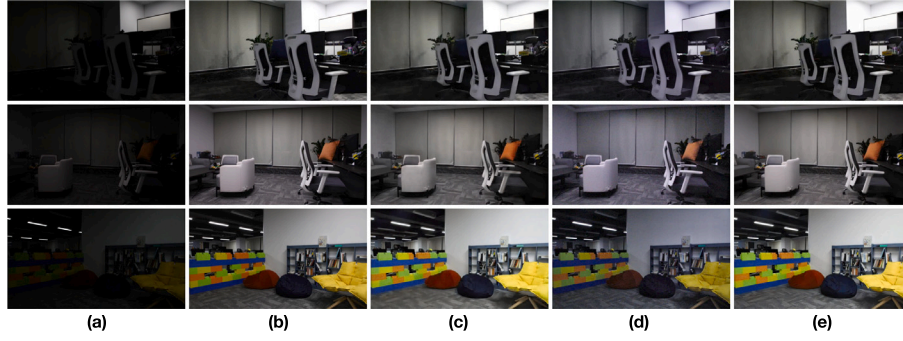


Fig. 5. Three images from SDSD-indoor test set were selected for comparison with different methods. (a) Input Image. (b) Ground Truth. (c) ZeroDCE++. (d) SNRNet. (e) ECAFormer.

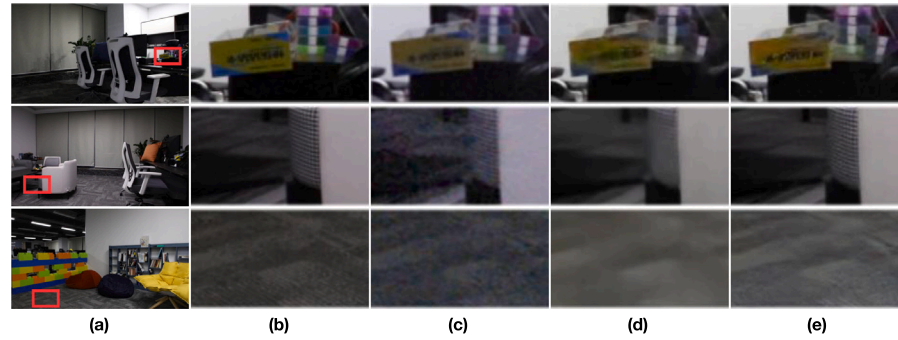


Fig. 6. The detailed comparison of the three images selected from SDSD-indoor test set. (a) Indicator. (b) Ground Truth. (c) ZeroDCE++. (d) SNRNet. (e) ECAFormer.

cropping 256×256 patches from pairs of low-/normal-light images. The batch size was set to 8. The training data was augmented with random rotations and flips to enhance variability and robustness. The training objective was to minimize both $\mathcal{L}_p$ 3.3.1 and $\mathcal{L}_c$ 3.3.2 between the enhanced images and their corresponding ground truths, ensuring high fidelity in the restoration of image details and color accuracy. To evaluate the performance, we employ the peak signal-to-noise ratio (PSNR) and structural similarity index (SSIM) (Wang et al., 2004) as the primary metrics for evaluation.

4.1.3. Comparison methods

To validate the effectiveness of our derived ECAFormer, we conducted extensive comparisons with several state-of-the-art conventional, two branch and guided LLIE approaches. For the conventional LLIE, we compared the UFormer(UF) (Wang et al., 2022), EnlightenGAN(EnGAN) (Jiang et al., 2021), Restormer(RSTM) (Zamir et al., 2022), LLFormer(LLF) (Wang et al., 2023), and MIRNet (Zamir et al., 2020). For the two-branch LLIE, we compared the RetinexNet(RN) (Wei et al., 2018) and KinD (Zhang et al., 2019). For the guided LLIE, we compared the RUAS (Liu et al., 2021), DRBN (Yang et al., 2021a), RetinexFormer(RF) (Cai et al., 2023), U2E-Net (Khan et al., 2024) and SNR-Net (Xu et al., 2022)

4.2. Results analysis

4.2.1. Quantitative results

To validate the quantitative performance of ECAFormer, we compared our model across five public datasets using two metrics, PSNR and SSIM, with the results presented respectively in Tables 1 and 3. The results on our collected *Traffic-297* dataset are shown in Table 4. Specifically, our ECAFormer achieves a 3.7% improvement in PSNR and nearly a 1.3% improvement in SSIM over the suboptimal method. The results indicate that our model achieves competitive performance while maintaining a relatively low parameter count and computational complexity. Experimental results show that processing

Table 3

Comparison of SSIM of different methods across 5 public datasets.

Methods	LOL-v1	LOLv2-r	LOLv2-s	SID	SMID	AVG	Params
RN(BMVC 2018)	0.560	0.567	0.798	0.578	0.684	0.6374	0.84
KinD(ACMMM 2019)	0.790	0.641	0.578	0.583	0.634	0.6452	8.02
EnGAN(TIP 2021)	0.650	0.617	0.734	0.543	0.674	0.6436	114.35
RUAS(CVPR 2021)	0.720	0.723	0.652	0.581	0.744	0.6840	0.003
DRBN(TIP 2021)	0.830	0.831	0.927	0.577	0.781	0.7892	5.27
UF(CVPR 2022)	0.771	0.771	0.871	0.577	0.792	0.7564	5.29
RSTM(CVPR 2022)	0.823	0.827	0.830	0.649	0.758	0.7774	26.13
MIRNet(TPAMI 2022)	0.830	0.820	0.876	0.605	0.762	0.7786	31.76
SNR(CVPR 2022)	0.842	<u>0.849</u>	0.928	0.625	0.805	<u>0.8098</u>	4.01
LLF(AAAI 2023)	0.816	0.792	0.909	\	\	\	24.55
RF(ICCV 2023)	0.816	0.814	<u>0.930</u>	0.673	0.815	0.8096	1.61
U2E-Net(PR 2024)	<u>0.847</u>	\	\	\	\	\	\
ECAFormer	0.850	0.853	0.931	<u>0.660</u>	<u>0.810</u>	0.8208	2.50

Table 4

Comparison of PSNR and SSIM on the traffic-297 dataset.

	RN	ZeroDCE++	EnGAN	SNR	RF	ECAFormer
PSNR	18.56	18.35	18.19	30.06	<u>32.17</u>	33.26
SSIM	0.770	0.884	0.623	0.965	<u>0.975</u>	0.978

the visual and semantic features through the DMSA module allows for preserving detail information while ensuring consistent brightness. Transformer-based models such as UFormer, RetinexFormer, and Restormer primarily rely on global information extraction, which limits their ability to preserve local details. The introduction of shallow visual features in our approach effectively addresses the limitations of transformer-based models in preserving fine-grained, per-pixel details.

4.2.2. Quality results

To validate the quality performance of ECAFormer, we present the visual comparison of the results, as shown in Figs. 5–8. As we can observe, the recent fully supervised learning method, SNRNet, did



Fig. 7. The experimental results across different scenes. From the first row to the third row are (1) Input Image. (2) ECAFormer. (3) Ground Truth.

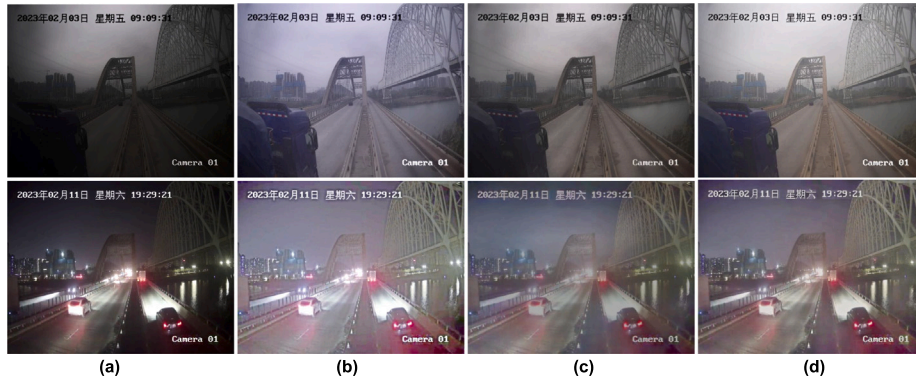


Fig. 8. Visual detail comparison on the Traffic-297. (a) Input Image. (b) ZeroDCE++. (c) SNRNet. (d) ECAFormer.

not retain sufficient detail and exhibited blurring. ZeroDCE++, which adjusts the pixel curves, introduced excessive noise. In the detailed comparison illustrated in Fig. 6, our method effectively preserves texture details while considering global brightness balance, indicating its ability to retain details and suppress noise. In Fig. 7, we showcase the performance of our model across different scenes, including color discrepancies caused by nighttime shooting. Specifically, our method was tested in complex scenarios from SDSD-in Wang et al. (2021), SDSD-out, and SMID datasets, with results indicating strong performance of ECAFormer. In Fig. 8, we evaluated the performance in traffic scenes, specifically testing models trained on synthesized datasets under real-world nighttime scenarios. The complexity of lighting conditions in nighttime traffic scenes poses a significant challenge to models. While other methods may introduce noise and lose details, our model demonstrates exceptional visual performance by effectively preserving details while considering global illumination. In comparison, ZeroDCE++ exhibits overexposure issues, highlighting the superior visual results of our proposed model. These results indicate that our model adapts well to variations and produces high-quality outcomes closely approximating the ground truth, demonstrating the effectiveness of ECAFormer as proposed in this paper.

Additionally, we conducted a user study on the LOL-v1 test set to identify the most visually pleasing enhancement results. We invited 50 volunteers to select the best result for each image in the test set. The results are shown in Fig. 9.

4.3. Ablation study

To assess the efficacy of the different modules, including (1) Visual-Semantic Feature (VSF): remove the visual semantic module and directly pass the input to DMSA., (2) Loss: only use L_1 loss without perceptual loss and charbonnier loss, and (3) DMSA: replace DMSA with conventional MHSA, we made adjustments to the model structure

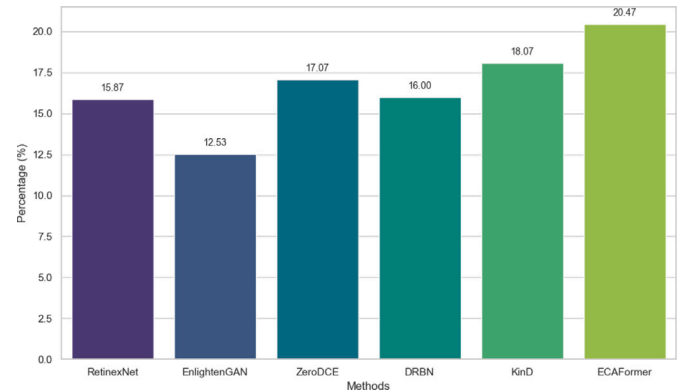


Fig. 9. User study on the test dataset of LOL-v1.

and presented the PSNR results in given Table 5 and Figs. 10–11. The results presented in Table 5 suggest that all three components positively impact the model's performance and are effective. Among them, the DMSA module shows the most significant improvement in feature fusion.

Furthermore, in ECAFormer, we utilize the DMSA module to effectively manage cross-layer residual information, thereby minimizing the loss of details resulting from downsampling. This validates the capability of the CSDMSA module to integrate residual information, as depicted in Fig. 10. It is evident that without the CSDMSA module, the enhanced images suffer from blurring and significant loss of image detail. This outcome indicates that the CSDMSA module effectively mitigates the loss of details caused by downsampling by utilizing cross-layer residual information.



Fig. 10. The ablation experiment of CSDMSA. Without the CSDMSA module, the images become blurred.

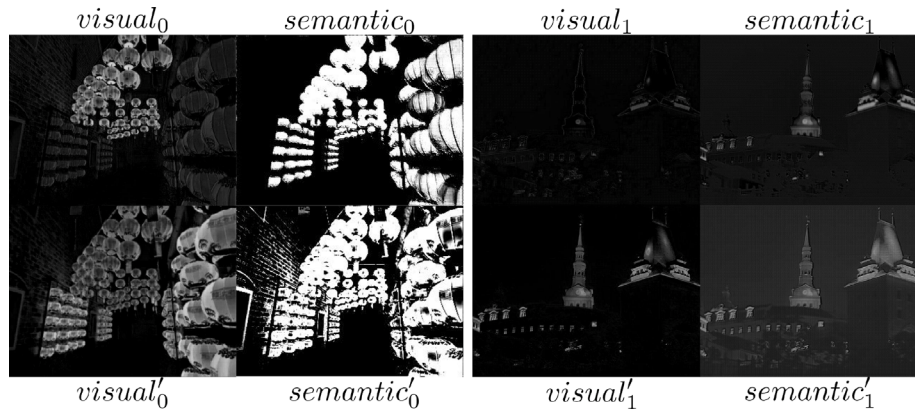


Fig. 11. Visualization of the two features before and after interaction through DMSA. The first row shows before interaction, and the second row shows after interaction.

Table 5
Ablation study results on SDSD-out.

VSF	Loss	DMSA	PSNR(avg)
✓			28.44
	✓		27.98
		✓	28.49
	✓	✓	28.56
✓		✓	29.41
✓	✓		28.38
✓	✓	✓	29.57

Besides, as depicted in Fig. 11, DMSA effectively facilitates the seamless integration of visual and semantic features. The findings demonstrate that visual features complement semantic features by providing additional intricate details, while semantic features offer long-range dependencies for visual features, resulting in improved brightness uniformity. Therefore, our designed DSMA significantly enhances the complementary information of visual and semantic features.

5. Conclusion

In this paper, we propose a transformer-based method for LLIE. Specifically, we use DMSA blocks to enhance low-light images by enhancing the semantic and visual features decoupled by a CNN network with progressively enlarged receptive fields. This module is capable of enhancing the two input features while preserving their distinction

without blending them. Furthermore, the ECAFormer network combines the advantages of local detailed feature information extraction in CNNs and global information extraction in transformer networks, which effectively reveals the local and global information of the image. Thus, the detailed information can be preserved after enhancement. Extensive experimental results validate the competitive advantages of ECAFormer compared to other state-of-the-art algorithms. In our future research, we will focus on refining the training approach to enable the model to effectively utilize disparate datasets and integrate additional prior knowledge to enhance its feature extraction capabilities. Additionally, we plan to use multi-objective optimization techniques (e.g., MOGMO (Pandya et al., 2024b), MOEDO (Pandya et al., 2024a)) to optimize the decoupled features, enhancing the distinction between the two features and thereby improving the differentiation of visual and semantic features.

CRediT authorship contribution statement

Yudi Ruan: Writing – original draft, Validation, Investigation. **Hao Ma:** Methodology, Conceptualization. **Di Ma:** Writing – original draft. **Weikai Li:** Writing – review & editing, Conceptualization. **Xiao Wang:** Writing – review & editing, Conceptualization.

Declaration of competing interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence

our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript.

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Data availability

Data will be made available on request.

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